
Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 2
July 15, 2025

Exam Duration: 5pm - 7pm.

Student Name (print): AMY WENTZELL

Student ID: 8820998

This exam contains 10 pages (including this cover page) and 5 questions. The total number of possible points is 50. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units for applied problems.**
- **Clearly identify your answer for each problem.**

Question	Points	Score
1	10	10
2	10	10
3	10	10
4	10	10
5	10	10
Total:	50	47

Do not write in the table to the right.

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1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) The expression for the volume integral $\iiint_T f(x, y, z) dx dy dz$ when the coordinates are changed from (x, y, z) to some other coordinates (u, v, w) .

Solution.

$$2 \quad \iiint_T f(x, y, z) = \iiint f(x(u, v, w), y(u, v, w), z(u, v, w)) \frac{\partial(u, v, w)}{\partial(x, y, z)} du dv dw$$

□

- (b) The main equation for **Stokes' Theorem** for a vector field $\vec{F}$ in a surface S with a boundary curve C traversed such that S remains on the left of C .

Solution.

$$2 \quad \oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dA$$

□

- (c) An expression for a vector $\vec{N}$ that is normal to the surface $z - x^2 - y^2 = 0$. $z = x^2 + y^2$

Solution.

$$6 \quad \begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \\ z &= r^2 \end{aligned} \quad \therefore \vec{N} = [0, 0, 1] \quad X$$

$$\vec{N} = \nabla g = \begin{bmatrix} -2x \\ -2y \\ 1 \end{bmatrix}$$

□

- (d) The vector field $\vec{F}$ polar coordinates written as $F_r \hat{e}_r + F_\theta \hat{e}_\theta$ if in Cartesian coordinates we have $\vec{F} = F_1 \hat{i} + F_2 \hat{j} = (x^2 + y^2) \hat{i} - 2y \hat{j}$. Hint: $F_r = F_1 \cos \theta + F_2 \sin \theta$, $F_\theta = -F_1 \sin \theta + F_2 \cos \theta$.

Solution.

$$1 \quad \begin{aligned} \theta &= \tan^{-1}\left(\frac{y}{x}\right) \\ \sin \theta &= \frac{-2y}{\sqrt{(x^2+y^2)}} \\ \cos \theta &= \frac{(x^2+y^2)}{\sqrt{(x^2+y^2)}} \end{aligned}$$

$$F_r = F_1 \cos \theta + F_2 \sin \theta$$

$$F_\theta = -(x^2+y^2) \sin \theta - 2y \cos \theta$$

$$\therefore \vec{F} = [(x^2+y^2) \cos \theta - 2y \sin \theta] \hat{e}_r + [-(x^2+y^2) \sin \theta - 2y \cos \theta] \hat{e}_\theta$$

$$\text{use: } x^2+y^2 = r, y = r \sin \theta$$

- (e) The expression for the divergence $\nabla \cdot \vec{F}$ in spherical coordinates (ρ, θ, ϕ) if we are given that the divergence of the same vector field is $\nabla \cdot \vec{F} = r^2 + z^2$ in cylindrical coordinates (r, θ, ϕ) .

Solution.

$$2 \quad \therefore \nabla \cdot \vec{F} = r^2 + z^2 = \rho^2$$

□

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2. (10 points) Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function

$$f(x, y, z) = x^2 + y^2 + z^2$$

in both Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.

Solution.

Cylindrical

$$f(x, y, z) = r^2 + z^2, h_1 = 1, h_2 = r, h_3 = 1$$

$$\nabla^2 f = \frac{1}{r} \left[\frac{\partial}{\partial r} \left(r \frac{\partial (r^2 + z^2)}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial (r^2 + z^2)}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(r \frac{\partial (r^2 + z^2)}{\partial z} \right) \right]$$

$$\nabla^2 f = \frac{1}{r} \left[\frac{\partial}{\partial r} (r \cdot 2r) + \frac{\partial}{\partial z} (r \cdot 2z) \right]$$

$$\nabla^2 f = \frac{1}{r} \left[\frac{\partial}{\partial r} (2r^2) + \frac{\partial}{\partial z} (2rz) \right]$$

$$\nabla^2 f = \frac{1}{r} (4r + 2r)$$

$$\nabla^2 f = \frac{1}{r} (6r)$$

$$\therefore \nabla^2 f = 6$$

Spherical

$$f(x, y, z) = \rho^2$$

$$h_1 = 1, h_2 = \rho \sin \phi, h_3 = \rho$$

$$\nabla^2 f = \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} \left(\rho^2 \sin \phi \frac{\partial (\rho^2)}{\partial \rho} \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \phi} \frac{\partial (\rho^2)}{\partial \theta} \right) + \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial (\rho^2)}{\partial \phi} \right) \right]$$

$$\nabla^2 f = \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} (\rho^2 \sin \phi (2\rho)) \right]$$

$$\nabla^2 f = \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} (2\rho^3 \sin \phi) \right]$$

$$\nabla^2 f = \frac{1}{\rho^2 \sin \phi} (6\rho^2 \sin \phi)$$

$$\therefore \nabla^2 f = 6$$

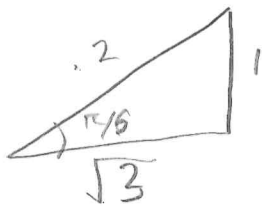
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3. (10 points) Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

Solution.



$$ds^2 = d\rho^2 + \rho^2 \sin^2 \phi d\theta^2 + \rho^2 d\phi^2 \rightarrow 0$$

$$ds^2 = d\rho^2 + \rho^2 \sin^2 \phi d\theta^2$$

$$ds = d\rho + \rho \sin \phi d\theta$$

$$\therefore dA = \rho \sin \phi d\rho d\theta$$

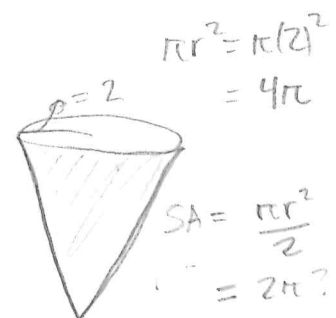
$$SA = \int_0^{2\pi} \int_0^2 \rho \sin\left(\frac{\pi}{6}\right) d\rho d\theta$$

$$SA = \int_0^{2\pi} \left[\frac{1}{2} \rho^2 \left(\frac{1}{2}\right) \right]_0^2 d\theta$$

$$SA = \int_0^{2\pi} \frac{1}{4} (2)^2 d\theta$$

$$SA = \int_0^{2\pi} d\theta$$

$$\therefore SA = 2\pi$$



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4. (10 points) Verify Gauss' Divergence Theorem by computing the following surface integral:

$$\oiint_S \vec{F} \cdot \hat{n} \, dA,$$

and the volume integral:

$$\iiint_T (\nabla \cdot \vec{F}) \, dV,$$

where $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$, S is the surface of a sphere of radius 2 that forms the boundary of T .Hint: This problem may be easier to solve in the spherical coordinates. Note that in spherical coordinates, $\vec{F} = \rho\hat{e}_\rho$, while for S , $\hat{n} = \hat{e}_\rho$, and thus

$$\vec{F} \cdot \hat{n} = (\rho\hat{e}_\rho) \cdot (\hat{e}_\rho) = \rho.$$

$$ds^2 = d\rho^2 + \rho^2 d\theta^2 + \rho^2 \sin^2\theta d\phi^2$$

$$dA = \rho^2 \sin\theta \, d\theta \, d\phi$$

Solution.

$$\vec{F} = [x, y, z],$$

$$\begin{aligned} \oiint_S \vec{F} \cdot \hat{n} \, dA &= \int_0^{2\pi} \int_0^\pi \rho (\rho^2 \sin\theta) \, d\theta \, d\phi \\ &= \int_0^{2\pi} \int_0^\pi \rho^3 \sin\theta \, d\theta \, d\phi \\ &= 8 \int_0^{2\pi} \left[-\cos\theta \right]_0^\pi \, d\phi \\ &= 8 \int_0^{2\pi} [-(-1) - (-1)] \, d\phi \\ &= 8 \int_0^{2\pi} 2 \, d\phi \\ &= 16 \theta \Big|_0^{2\pi} \end{aligned}$$

$$\therefore \oiint_S \vec{F} \cdot \hat{n} \, dA = 32\pi$$

□

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Continue problem 4

Solution.

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial(x)}{\partial x} + \frac{\partial(y)}{\partial y} + \frac{\partial(z)}{\partial z} = \underline{\underline{3}}$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi, \quad 0 \leq \rho \leq 2 \text{ for Volume}$$

$$\iiint_V (\vec{\nabla} \cdot \vec{F}) \, dV = 3 \int_0^{2\pi} \int_0^\pi \int_0^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$= 3 \int_0^{2\pi} \int_0^\pi \left[\frac{1}{3} \rho^3 \right]_0^2 \sin \phi \, d\phi \, d\theta$$

$$= 3 \int_0^{2\pi} \int_0^\pi \frac{8}{3} \sin \phi \, d\phi \, d\theta$$

$$= 8 \int_0^{2\pi} \left[-\cos \phi \right]_0^\pi d\theta \quad \begin{array}{l} \text{evaluated} \\ \text{already,} \\ \Rightarrow 2 \end{array}$$

$$= 8 \int_0^{2\pi} 2 \, d\theta$$

$$= 16 \, \theta \Big|_0^{2\pi}$$

$$\therefore \iiint_V (\vec{\nabla} \cdot \vec{F}) \, dV = 32\pi \quad \checkmark$$

□

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5. (10 points) In the first quadrant of the Cartesian plane (i.e., $x \geq 0, y \geq 0$), we may define a set of coordinates known as the hyperbolic coordinates (u, v) , with transformation function

$$x = ve^u, \quad y = ve^{-u}.$$

Find the area of the triangular region R given by

$$0 \leq v \leq u, \quad 0 \leq u \leq 3.$$

Solution.

$$h_u^2 = \left(\frac{\partial x}{\partial u}\right)^2 + \left(\frac{\partial y}{\partial u}\right)^2$$

$$h_u^2 = (ve^u)^2 + (-ve^{-u})^2$$

$$h_u^2 = v^2 e^{2u} + v^2 e^{-2u}$$

$$h_u^2 = v^2 (e^{2u} + e^{-2u})$$

$$h_u = v \sqrt{e^{2u} + e^{-2u}}$$

$$h_v^2 = \left(\frac{\partial x}{\partial v}\right)^2 + \left(\frac{\partial y}{\partial v}\right)^2$$

$$h_v^2 = (e^u)^2 + (e^{-u})^2$$

$$h_v^2 = e^{2u} + e^{-2u}$$

$$h_v = \sqrt{e^{2u} + e^{-2u}}$$

Use $\frac{\partial(x,y)}{\partial(u,v)} = |\det D| = 2v$

$$dA = h_u h_v du dv$$

$$SA = \int_0^3 \int_0^u v(e^{2u} + e^{-2u}) dv du$$

$$SA = \int_0^3 \frac{(e^{2u} + e^{-2u})}{2} v^2 \Big|_0^u du$$

$$SA = \frac{1}{2} \int_0^3 u^2 (e^{2u} + e^{-2u}) du$$

IBP: $\int u^2 (e^{2u} - e^{-2u}) du$

$$u = u^2 \quad dv = e^{2u} - e^{-2u}$$

$$du = 2u \quad v = \frac{1}{2}e^{2u} + \frac{1}{2}e^{-2u}$$

$$\Rightarrow = u^2 \left(\frac{1}{2}\right)(e^{2u} + e^{-2u}) - \int 2u \left(\frac{1}{2}\right)(e^{2u} + e^{-2u}) du$$

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5. IBP#2

$$\int u (e^{2u} + e^{-2u}) = u \left(\frac{1}{2} \right) (e^{2u} - e^{-2u}) - \int \left(\frac{1}{2} \right) (e^{2u} - e^{-2u}) du$$

$$u = u \\ du = 1$$

$$dv = e^{2u} + e^{-2u} \\ v = \frac{1}{2} e^{2u} - \frac{1}{2} e^{-2u}$$

$$SA = \frac{1}{2} \left\{ \frac{u^2}{2} (e^{2u} + e^{-2u}) - \left[\frac{u}{2} (e^{2u} - e^{-2u}) - \frac{1}{4} (e^{2u} + e^{-2u}) \right] \right\}_0^3$$

$$SA = \frac{1}{2} \left\{ \frac{9}{2} (e^6 + e^{-6}) - (0) - \left[\frac{3}{2} (e^6 - e^{-6}) - (0) - \frac{1}{4} (e^6 + e^{-6}) + \frac{1}{4} (1+1) \right] \right\}$$

$$SA = \frac{1}{2} \left\{ \frac{9}{2} (e^6 + e^{-6}) - \frac{3}{2} (e^6 - e^{-6}) + \frac{1}{4} (e^6 + e^{-6}) - \frac{1}{2} \right\}$$

$$SA = \frac{1}{2} \left[\frac{19}{4} (e^6 + e^{-6}) - \frac{3}{2} (e^6 - e^{-6}) - \frac{1}{2} \right]$$

$$SA = \frac{1}{2} \left[\left(\frac{19}{4} - \frac{3}{2} \right) e^6 + \left(\frac{19}{4} + \frac{3}{2} \right) e^{-6} - \frac{1}{2} \right]$$

$$\therefore SA = \frac{13}{8} e^6 + \frac{25}{8} e^{-6} - \frac{1}{4}$$

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Formula Sheet

General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

$$\text{Derivative Matrix: } D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$$

$$\text{Lamé coefficients: } h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

$$\text{Jacobian: } \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D| = |h_1 h_2 h_3|$$

$$\text{Line element: } ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

$$\text{Volume element: } dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$$

$$\text{Area element for surface with } q_j = \text{const} \ (i, j, k \text{ distinct}): dA = h_i h_k dq_i dq_k$$

$$\text{Gradient for scalar function } f: \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_{q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_{q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_{q_3}$$

$$\text{Divergence and curl of vector field } \vec{F} = F_{q_1} \hat{e}_{q_1} + F_{q_2} \hat{e}_{q_2} + F_{q_3} \hat{e}_{q_3}:$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2} (h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3} (h_1 h_2 F_{q_3}) \right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_{q_1} & h_2 \hat{e}_{q_2} & h_3 \hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Cylindrical Coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$